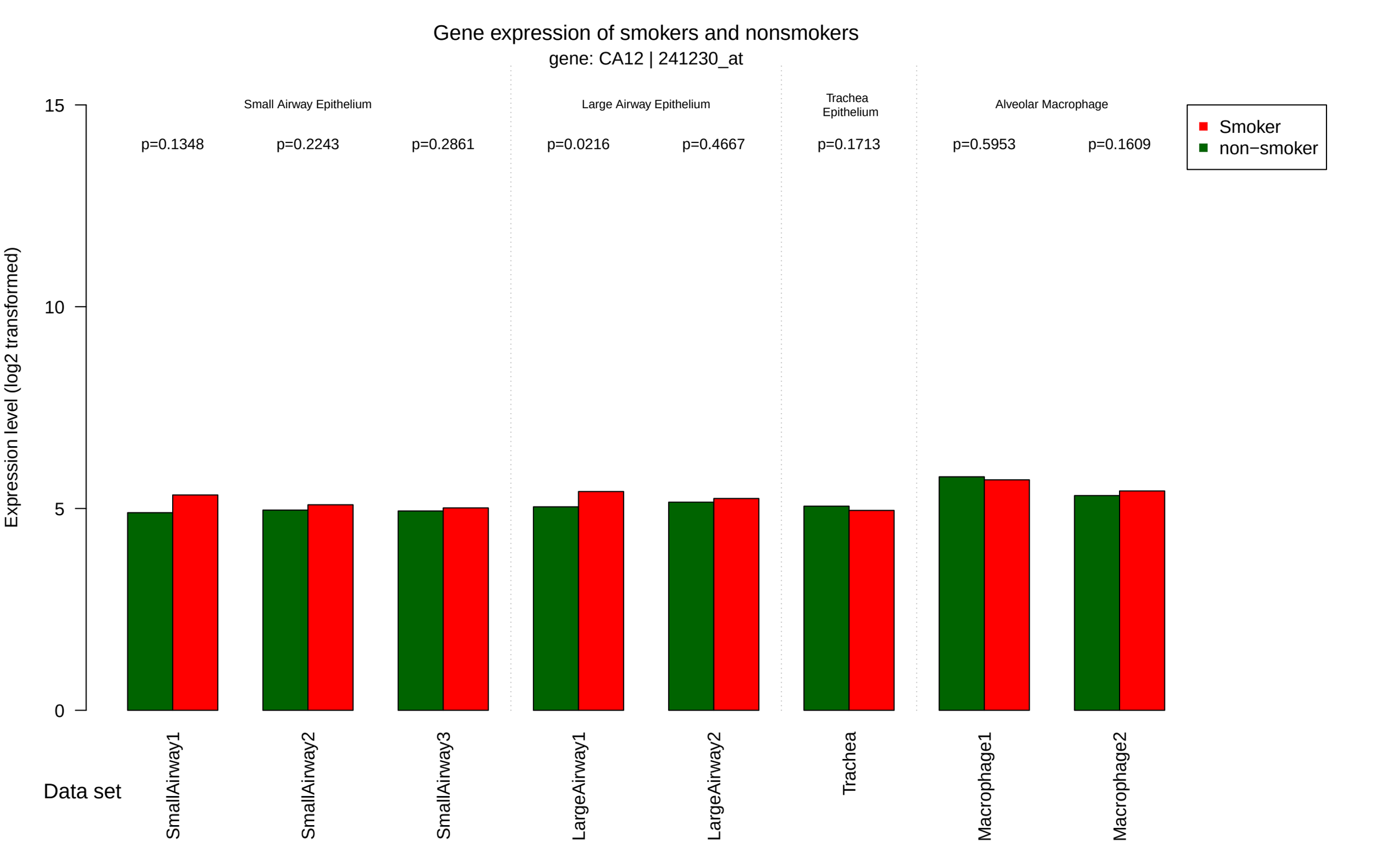
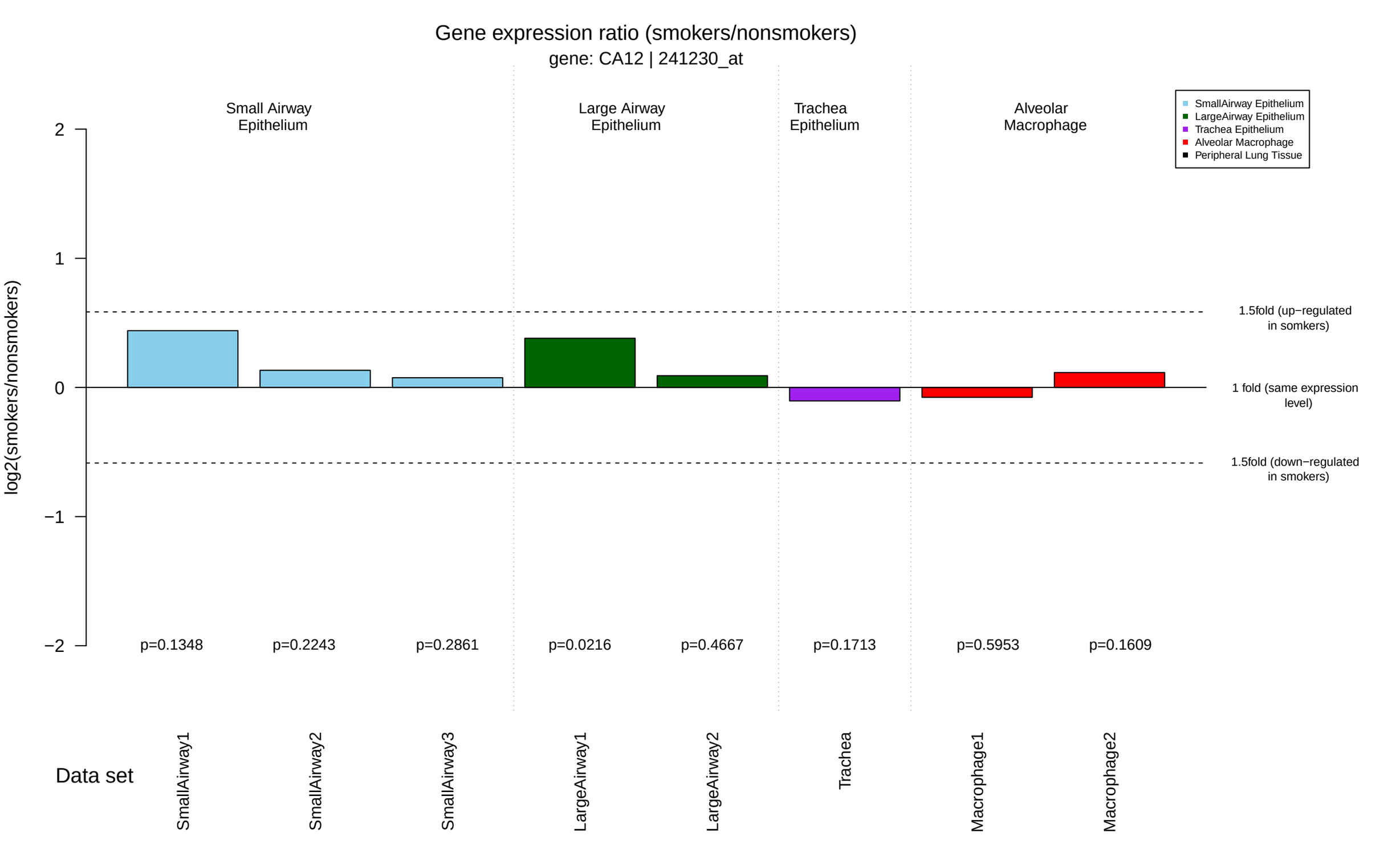
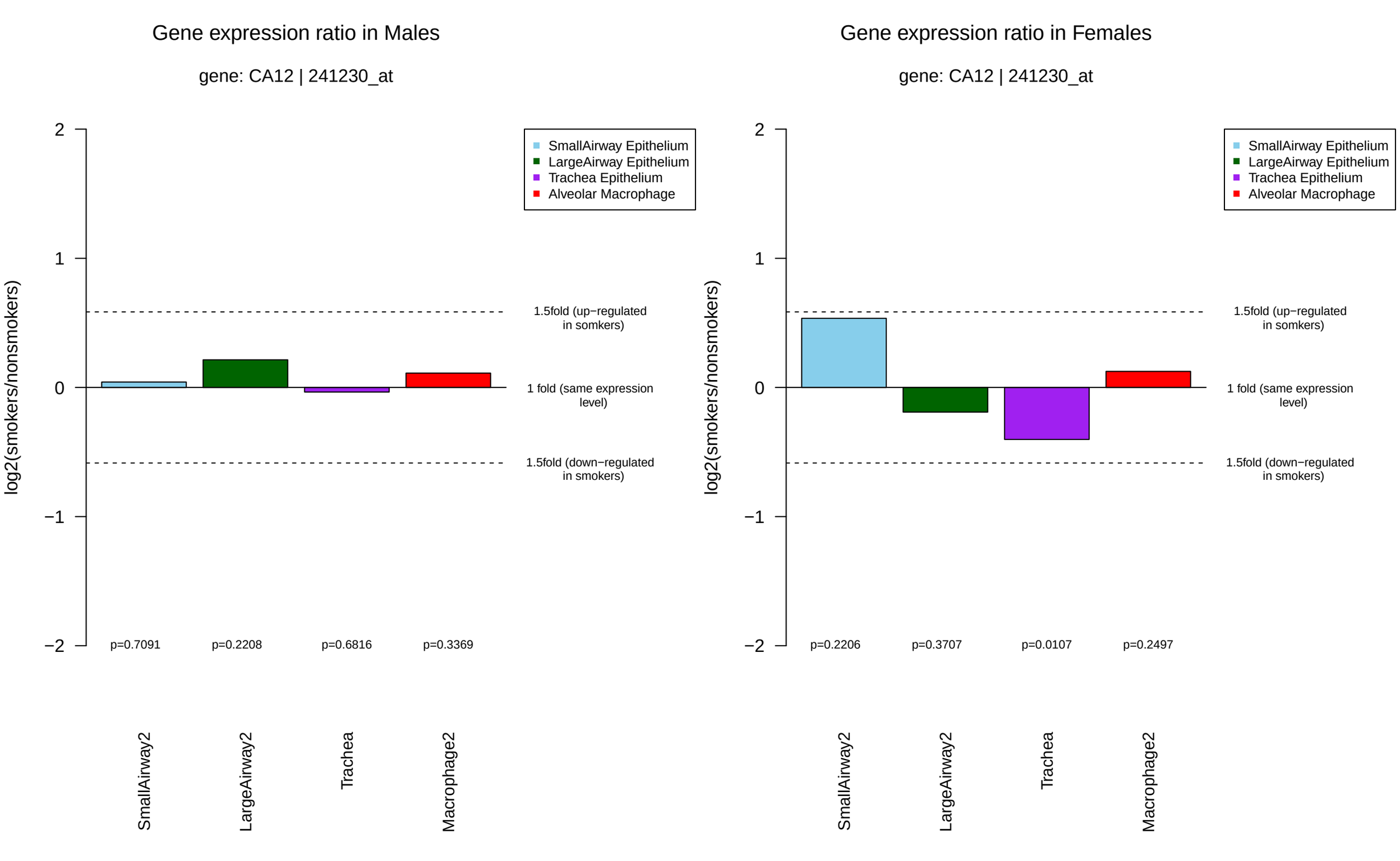
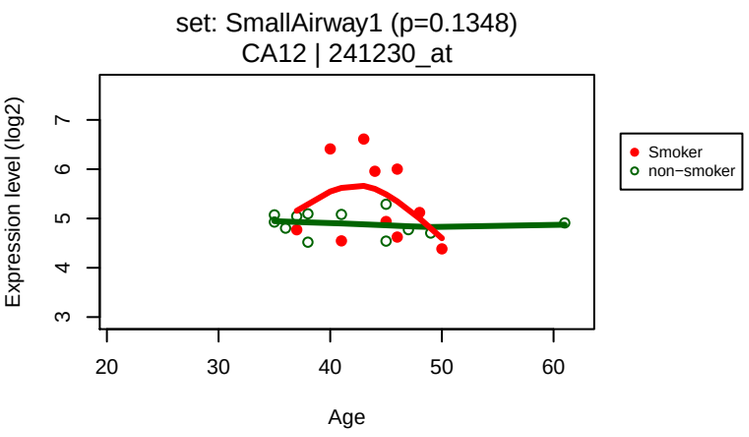
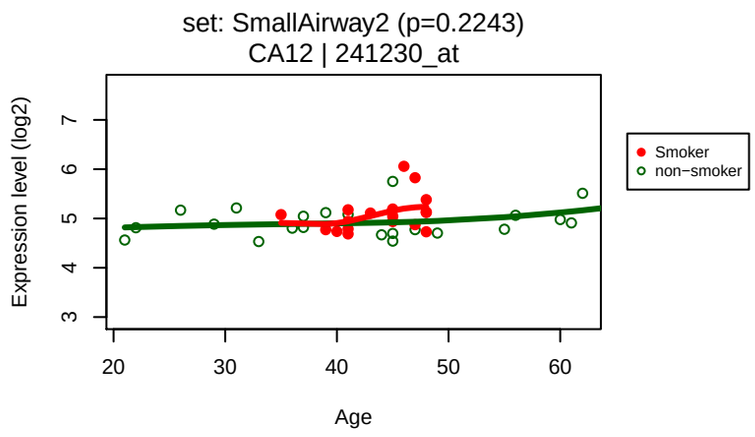
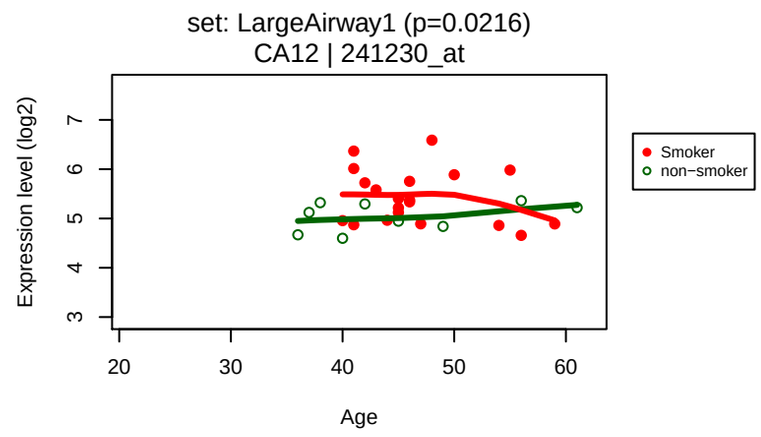

Gene expression of smokers and nonsmokers
gene: CA12 | 241230_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1348
p=0.2243
p=0.2861
p=0.0216
p=0.4667
p=0.1713
p=0.5953
p=0.1609
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

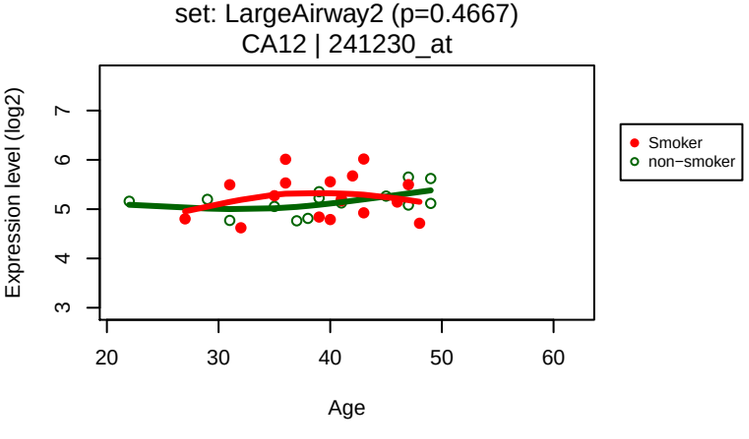

Gene expression ratio (smokers/nonsmokers)
gene: CA12 | 241230_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1348
p=0.2243
p=0.2861
p=0.0216
p=0.4667
p=0.1713
p=0.5953
p=0.1609
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

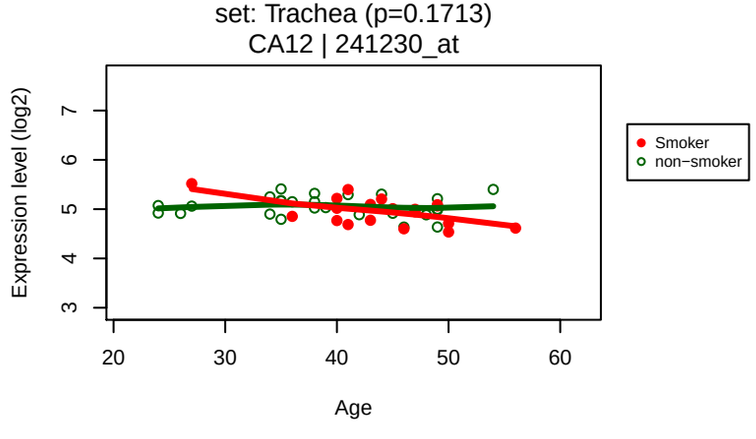

Gene expression ratio in Males
gene: CA12 | 241230_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7091
p=0.2208
p=0.6816
p=0.3369
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA12 | 241230_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2206
p=0.3707
p=0.0107
p=0.2497
SmallAirway2
LargeAirway2
Trachea
Macrophage2

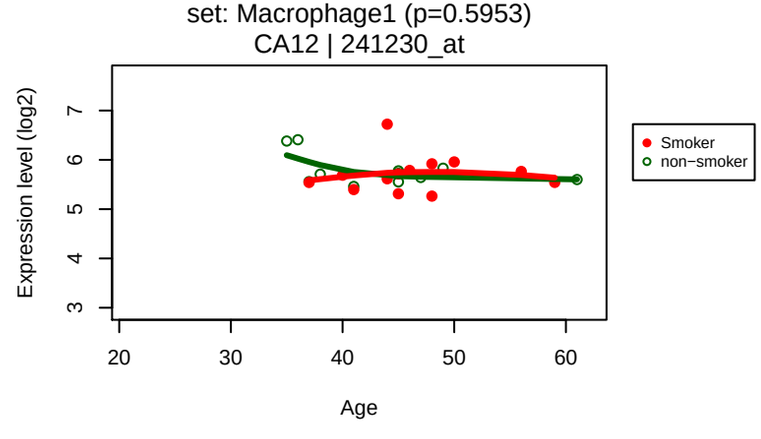

set: SmallAirway1 (p=0.1348)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker

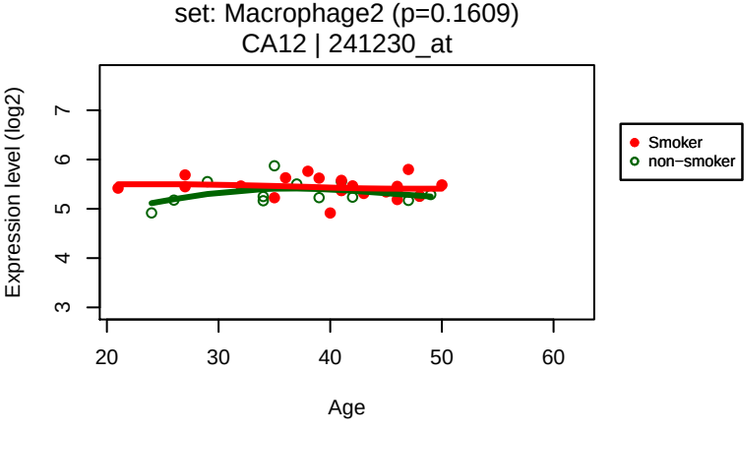

set: SmallAirway2 (p=0.2243)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker

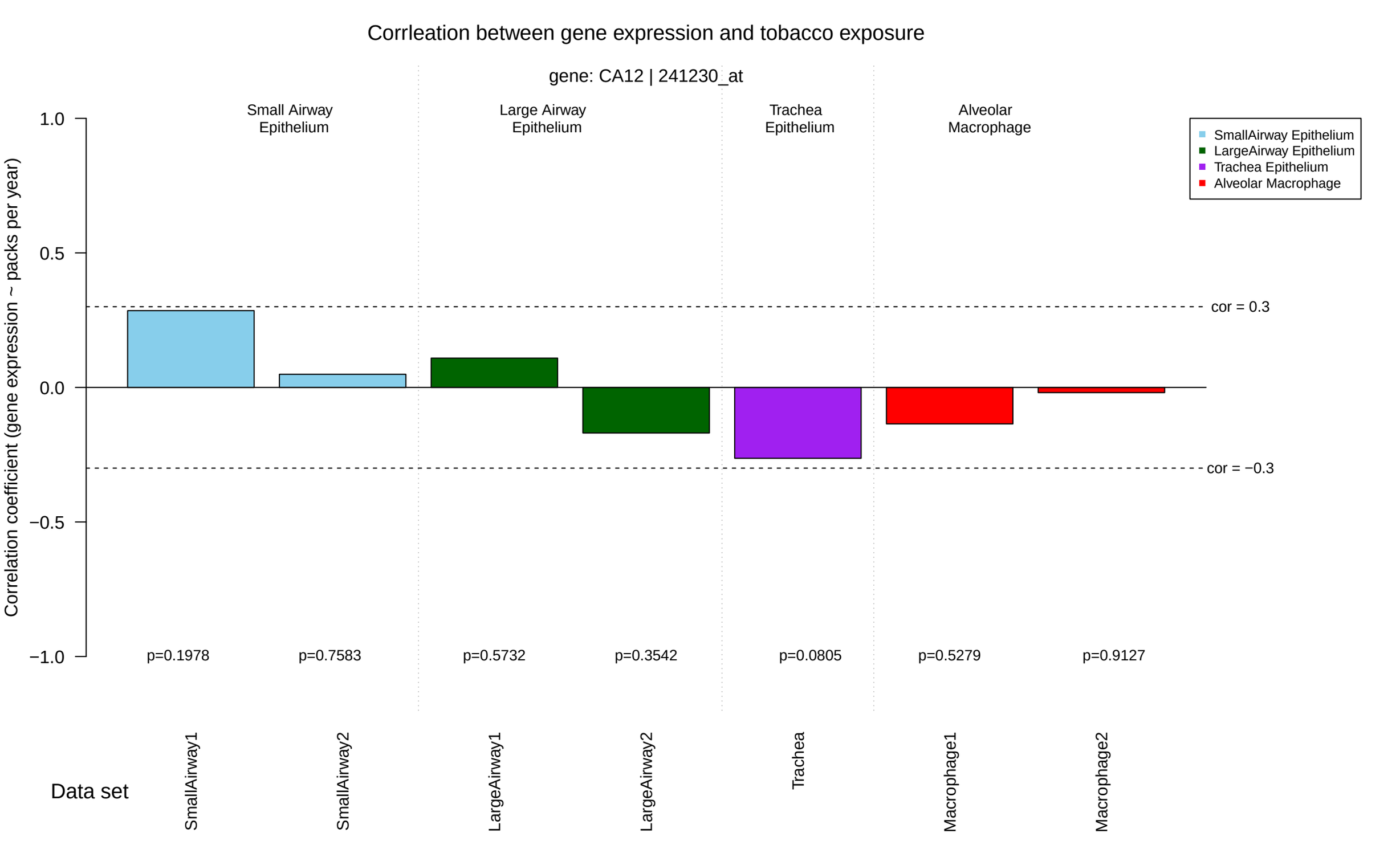

set: LargeAirway1 (p=0.0216)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4667)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1713)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5953)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1609)
CA12 | 241230_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA12 | 241230_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1978
p=0.7583
p=0.5732
p=0.3542
p=0.0805
p=0.5279
p=0.9127
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2